

TDKs new offerings

TDK announced four new speakers: Wireless Weatherproof Speaker; Wireless Boombox; Wireless SoundCube and Wireless Charging Speaker at IFA, Berlin

**Jarre Aeroskull**

No those aren't funky shade-holders, but a cool speaker designed by Jarre Technologies

AUDIOPHILIA: Types of speakers

We give you the basic soundbytes on speakers and their types. Which one resonates best with you?

Nimish Sawant

nimish.sawant@thinkdigit.com

We all have seen speakers evolve over time. Back in the early 90s, as kids, we were blown away with the cupboard-sized speakers that we saw in the Micheal Jackson hit *Black or White*. In this day and age, speakers have shed their bulk. Where back in the day the audio sources were equally hefty, the audio source for which current generation speakers are designed can fit in your pocket.

So what really is the science behind these everyday instruments? What decides a speaker's design? We explore various types of speakers and how they work. And also get back to the age-old form vs function debate.

1 HORN SPEAKERS

Remember the logo of the famous recording label called HMV? It had a dog peering into a triangular thing. It is the same thing that you see atop those old gramophones. It is called the Horn Speaker, one of the earliest form of loud speakers. Horn speakers are passive components and do not use electricity. The thinner part of the horn near the driver element is called the 'throat' and the wide opening at the other end is called the 'mouth'. Mechanical gramophones used them as early as 1857, but nowadays you do get electric horn speakers as well.



Due to the conical design of the horn speakers, the horn speaker does not amplify sound from the driver element but tends to improve the coupling efficiency between the driver and the air. The smaller diameter high pressure condition at the throat of the horn speaker is converted to a larger diameter low pressure condition at the mouth. This increases the sensitivity of the speaker thereby leading to an increase in the volume of the sound. To get a more bass-heavy sound, the length of the horn mattered in direct proportion.

2 ELECTRODYNAMIC LOUDSPEAKER

This is the loudspeaker that is most commonly used in the speakers all around us. It uses an electromagnetic coil and a diaphragm to create sound. It comprises four main parts – a stationary magnet, a coil of copper wires, a suspender spider and a cardboard/paper cone which acts as a diaphragm which is suspended. As you pass current through the copper coil, also known as the voice coil, electricity is generated, which interacts with the magnetic

field of the permanent magnet to cause displacement. This displacement causes the suspender spider and in turn the diaphragm to vibrate thereby producing sound. A particular speaker is designed to produce a specific range of frequency. Different diaphragm materials will produce different kind of sounds.

Good quality speakers have a woofer, a mid-range driver and a tweeter included within an enclosure. Each of these sub-parts can cater to different frequency bands – the mid-range tweeters take care of mid-range frequencies, woofers take care of low frequencies and tweeters are concerned with higher frequencies. Using an electrical circuit known as a crossover which comprises inductors, capacitors among other electrical components, the signal coming from the source can be split up into various



frequencies – low, mid, high – and each of these frequencies can be channeled to the respective speakers which are optimised to output them. As for the enclosure, it's a very vast topic and perhaps deserves a separate write-up in itself.

3 WIRELESS SPEAKERS

They are very similar to normal speakers except for the fact that they transmit audio signals using radio frequency (RF) waves instead of using cables. The wireless speakers comprise